

Fibonacci Numbers

Time limit: 1.00 s

Memory limit: 512 MB

The Fibonacci numbers can be defined as follows:

- $F_0 = 0$
- $F_1 = 1$
- $F_n = F_{n-2} + F_{n-1}$

Your task is to calculate the value of F_n for a given n .

Input

The only input line has an integer n .

Output

Print the value of F_n modulo $10^9 + 7$.

Constraints

- $0 \leq n \leq 10^{18}$

Example

Input	Output
10	55